Extra slides

© Copyright 2004-2026, Bootlin
Creative Commons BY-SA 3.0
Corrections, suggestions, contributions and translations are welcome!





Quilt



Overview

- Quilt is a utility to manage patches which can be used without having a clean source tree.
- It can be used to create patches for recipes already available in the build system.
- Be careful when using this workflow: the modifications won't persist across builds!



Using Quilt

1. Find the recipe working directory in `$BUILDDIR/tmp/work/`.
2. Create a new `Quilt` patch: `$ quilt new topic.patch`
3. Add files to this patch: `$ quilt add file0.c file1.c`
4. Make the modifications by editing the files.
5. Test the modifications: `$ bitbake -c compile -f recipe`
6. Generate the patch file: `$ quilt refresh`
7. Move the generated patch into the recipe's directory.



Runtime Package Management (IPK Setup)



Requirements

- First of all, you need a server to serve the packages to a private subnet or over the Internet. Packages are typically served over `https` or `http`.
- Specific tools are also required on the target, and must be shipped on the product. They should be included into the images generated by the build system.
- These tools will be specific to the package type used.
  - This is similar to Linux distributions: Debian is using `.deb` related tools (dpkg, apt…) while Fedora uses `.rpm` related ones (rpm, dnf).



Build configuration 1/2

- The PACKAGE_CLASSES variable controls which package format to use. More than one can be used.
- Valid values are `package_rpm`, `package_deb`, `package_ipk`.
- By default Poky uses the RPM format, while OpenEmbedded-Core uses the IPK one.
- Example:
  - `PACKAGE_CLASSES = "package_ipk"`
  - `PACKAGE_CLASSES = "package_rpm package_deb"`



Build configuration 2/2

To install the required tools on the target, there are two possible solutions:

- By adding `package-management` to the images features.
  - The required tool will be installed on the target.
  - The package database corresponding to the build will be installed as well.
- Or by manually adding the required tools in IMAGE_INSTALL. For example, to use the IPK format we need `opkg`.



Build considerations

- The Runtime Package Management uses package databases to store information about available packages and their version.
- Whenever a build generates a new package or modifies an existing one, the package database must be updated.
- `$ bitbake package-index`
- Be careful: BitBake does not properly schedule the `package-index` target. You must use this target alone to have a consistent package database.
  - `$ bitbake ninvaders package-index` won't necessarily generate an updated package database.



Package server configuration: Apache2 example setup

Apache2 HTTP setup for IPK packages. This should go in `/etc/apache2/sites-enabled/package-server.conf`.

```
<VirtualHost *:80>
    ServerName packages.example.net

    DocumentRoot /path/to/build/tmp/deploy/ipk
    <Directory /path/to/build/tmp/deploy/ipk>
        Options +Indexes
        Options Indexes FollowSymLinks
        Order allow,deny
        allow from all
        AllowOverride None
        Require all granted
    </Directory>
</VirtualHost>
```



Target configuration: configuring opkg

- The IPK runtime management software is `opkg`.
- It can be configured using configurations files ending in `.conf` in `/etc/opkg/`.
- This configuration helps `opkg` to find the package databases you want to use.
- For example, with our previously configured package server:

```
src/gz all http://packages.example.net/all
src/gz armv7a http://packages.example.net/armv7a
src/gz beaglebone http://packages.example.net/beaglebone
```

- This can be automatically generated by defining the PACKAGE_FEED_URIS, PACKAGE_FEED_BASE_PATHS and PACKAGE_FEED_ARCHS variables



opkg usage

- `opkg update`: fetch and update the package databases, from the remote package servers.
- `opkg list`: list available packages.
- `opkg upgrade`: upgrade all installed packages.
- `opkg upgrade <package>`: upgrade one package explicitly.
- `opkg install <package>`: install a specific package.



`opkg` upgrade over an unstable network

- To avoid upgrade issues when downloading packages from a remote package server using an unstable connection, you can first download the packages and then proceed with the upgrade.
- To do this we must use a cache, which can be defined in the `opkg` configuration with: `option cache /tmp/opkg-cache`.

```
# opkg update
# opkg --download-only upgrade
# opkg upgrade
```